

# Pre-analysis of task\_003: the task, the controller defect, and a coarse-graining that makes information quantities estimable

MA-CC blackboard-control study

2 October 2026

## Abstract

This report consolidates an offline re-analysis of the first batch of blackboard-control simulations (study 21-09-2026-full-vs-report-v1, 60 initializations of `task_003` with `gpt-oss-120b`). It reaches three conclusions. First, `task_003` is fully reconstructible from its nine hidden latent values, and the population of agents collectively holds the only short proof of the correct answer. Second, the controller used in that batch is *not* what its configuration claims: fact admission was evaluated once for the decoy target and reused for both steering directions, so the truth-targeting arm recommends `ALLOCATION_0` while quoting evidence selected to undermine it. Measured target information is 0.022 bits — essentially zero. Third, the raw population state (a 325-valued vote vector) defeats every estimator we tried; binning the mean agent posterior into four states makes the path divergence *exactly* computable, but four independent tests agree the resulting chain is Markovian only in the silent arm at persistence  $\rho = 1$ . No simulations were run and no provider requests made.

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## 1 Scope and vocabulary

Three terms recur and are defined once here.

**Blackboard.** A shared message board. Each round, every agent may post a statement; all posts are visible to all agents in that round, and the board is cleared before the next round. Nothing persists on the board itself.

**Epistemic persistence  $\rho$ .** Each agent remembers each fact it holds with probability  $\rho$ , independently per agent, per fact, per round. At  $\rho = 1$  memory is perfect; at  $\rho = 0.75$  a quarter of each agent’s knowledge decays every round and must be re-acquired from the board to survive.

**Controller.** An extra participant that may post selected true facts to steer the population toward a named allocation (its *target*). It is *silent* in the baseline arm. In the intervention arms its target is either `ALLOCATION_0` (the correct answer, called the *truth* direction) or `ALLOCATION_2` (an incorrect answer, the *decoy* direction).

## 2 What task\_003 is

### 2.1 The underlying problem

The task is a Team Allocation problem in the style of MuSR. Three people must be split between two jobs: one person works alone on the first job, the remaining two work jointly on the second. There are therefore exactly three candidate allocations, one per choice of the lone worker.

Nine hidden integers in  $\{1, 2, 3\}$  describe the world: for each of the three people, a skill level at each of the two jobs (six numbers), plus a cooperation level for each of the three possible pairs (three numbers). An allocation’s score is

$$\text{score} = \underbrace{s_{\text{singleton},1}}_{\text{lone worker's skill at job 1}} + \underbrace{\sum_{p \in \text{pair}} s_{p,2}}_{\text{pair's skills at job 2}} + \underbrace{c_{\text{pair}}}_{\text{how well the pair cooperates}}. \quad (1)$$

The highest-scoring allocation is the correct answer.

### 2.2 The specific world

`task_003` is the latent vector

$$V = (3, 1, 1, 2, 2, 1, 1, 1, 2), \quad (2)$$

reading the six skills first (person by person, job by job) then the three cooperation values. Scoring gives `ALLOCATION_0` as the unique winner. We rebuilt the task locally from  $V$  alone and verified the result is bit-identical to the archived frozen task: the cluster copies are not needed to reproduce anything in this report.

### 2.3 Facts

A *fact* is a true natural-language proposition about the world, drawn from a fixed catalog of 99 propositions: 45 *threshold* statements (“person  $p$  has at least moderate skill at job  $j$ ”) and 54 *comparison* statements (“ $p$  is better than  $p'$  at job  $j$ ”). Of these, **49 are true in task\_003** and only those can ever be posted. Nobody lies in these simulations; steering is done purely by *which* true facts get said.

## 2.4 Exact posterior

Because the world is only nine ternary numbers, we enumerate all  $3^9 = 19,683$  candidate worlds, keep the 14,388 with a unique winner, and for any set  $S$  of facts compute the posterior exactly by filtering:

$$\mathbb{P}(\text{ALLOCATION\_k} \mid S) = \frac{\#\{w : w \models S, \text{winner}(w) = k\}}{\#\{w : w \models S\}}. \quad (3)$$

This is a count, not an estimate. It is the ground truth against which every agent’s belief is scored. Implementation: `10_task_and_facts/engine.py`.

## 2.5 Who can prove the answer

A set  $S$  *solves* the task if every surviving world has `ALLOCATION_0` as winner, i.e.  $\mathbb{P}(\text{ALLOCATION\_0} \mid S) = 1$ . Enumerating minimal solving sets:

| quantity                                          | value          |
|---------------------------------------------------|----------------|
| smallest solving set                              | <b>4 facts</b> |
| number of 4-fact proofs                           | 24             |
| minimal proofs up to size 6                       | 7,694          |
| minimal proofs the agents can jointly assemble    | <b>4</b>       |
| minimal proofs the controller’s pool can assemble | <b>0</b>       |
| facts held by nobody                              | 12             |

The fact `cf_x00_eq_3` appears in *every* 4-fact proof and is in nobody’s starting hand. The population therefore holds the only short route to certainty, and barely: four proofs out of 7,694.

**Private-eligibility.** A fact is eligible to be handed out privately when it is individually weak:  $\max_k \mathbb{P}(\text{ALLOCATION\_k} \mid \{f\}) \leq 0.45$  *and* its normalized posterior entropy is  $\geq 0.90$ . Both conditions must hold. 41 of the 49 facts qualify; the remaining 8 are too revealing to distribute and are given to no one.

**The decisive set.** A greedy search over the 41 private-eligible facts produces a 6-fact set reaching  $\mathbb{P}(\text{ALLOCATION\_0}) = 1$ . These six are the *decisive* facts. They are a *subset* of the private-eligible ones — the two categories are nested, not disjoint, which is easy to double-count.

## 3 The controller defect

### 3.1 What went wrong

Admission of a fact into the controller’s pool is decided by whether it moves the posterior toward the target,  $\Delta\mathbb{P}(\text{target} \mid f) > 0$ , together with a non-decisiveness condition. In the archived batch this test was evaluated **once, for the decoy target** `ALLOCATION_2`, and the resulting 24-fact pool was reused for both configurations. Excluding decisive facts removes only 3 of them, all `ALLOCATION_2`-favouring.

The consequence is stark: in the truth arm the controller *recommends* `ALLOCATION_0` while *citing* facts chosen to push the population toward `ALLOCATION_2`. The two arms differ mainly in how often the controller fires at all — 32% of rounds in the truth arm against 54% in the decoy arm — not in what they say.

**Standing caveat.** The `ALLOCATION_0` arm is confounded. Its numbers must not be read as evidence about truth-directed steering. Its recorded *dynamics* are sound and are used wherever only dynamics matter.

### 3.2 The replacement

A corrected design uses two *matched symmetric pools*: 12 facts each, disjoint, built by running the admission test separately per target, and verified so that neither pool can prove its own target (otherwise the controller would simply hand over the answer). These are stored in `10_task_and_facts/symmetric_pools.json`.

## 4 Coarse-graining the population state

### 4.1 Why the raw state fails

The natural state — the vector of which allocation each agent currently favours — takes 325 values in these runs. With 1,800 to 5,000 transitions per cell, most of those states are visited once or never. Critic-based (variational) estimators of the path divergence on this space failed every reliability check we applied: the estimates did not stabilize with more data and their confidence intervals did not cover their own point estimates.

### 4.2 The epistemic coordinate

We replace the vote vector with a single number: the average correctness of belief across the population,

$$e_t = \frac{1}{N} \sum_{j=1}^N \mathbb{P}(\text{ALLOCATION\_0} \mid K_j(t)), \quad (4)$$

where  $K_j(t)$  is the set of facts agent  $j$  currently remembers and the posterior is the exact count of Eq. (3). So  $e_t$  is not a vote share and not an opinion average; it is “how close to the truth the population’s evidence actually puts it,” measured on the exact posterior.

We then bin  $e_t$  into **four states** with cut points at 0.33, 0.50, 0.75. The cuts are semantic rather than data-driven: 1/3 is the prior (knowing nothing), 0.5 is a bare majority of surviving worlds, and 0.75 is near-certainty. A sweep over 2–6 bins and over quantile versus semantic cuts (Fig. 4) shows 3, 4 and 5 bins are statistically indistinguishable once resampled across 12 seeds; 4 was chosen for data budget. An 8-state version that also tracks knowledge *spread* repairs the  $\rho = 1$  cells but leaves three rows of the transition matrix estimated from too few transitions, so 4 states are used everywhere.

**Never pool persistence levels.** The binning and the fitted kernel are estimated separately for each  $\rho$  and each arm.

### 4.3 Why this buys exactness

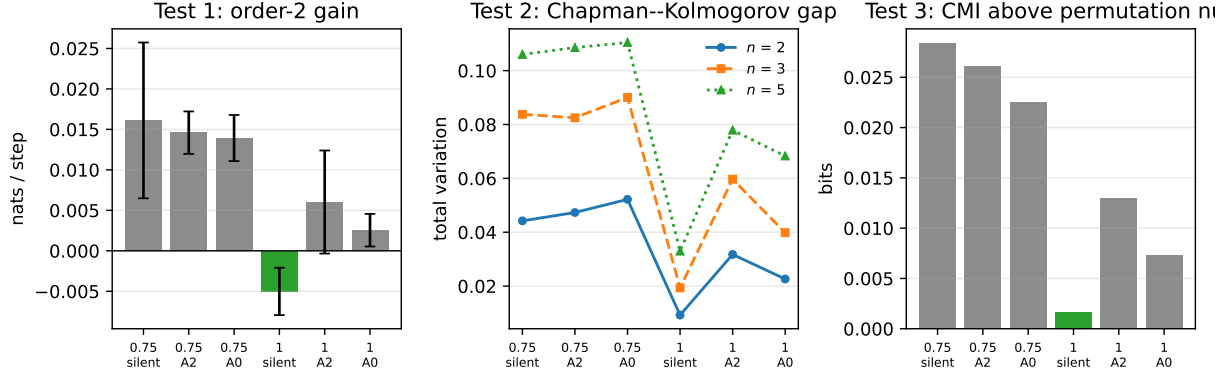
On a finite state space the divergence between two *trajectory laws* has a closed form. If  $P$  has initial distribution  $p_0$  and kernel  $T^P$ , and  $Q$  has  $q_0, T^Q$ , then over  $h$  steps

$$D(P \parallel Q) = D(p_0 \parallel q_0) + \sum_{t=0}^{h-1} \sum_i \mu_t(i) D(T^P(\cdot \mid i) \parallel T^Q(\cdot \mid i)), \quad (5)$$

with  $\mu_t$  the occupancy of  $P$  at time  $t$ . Every term is a finite sum over four states. **There is no critic and no variational bound** — that is the entire point of the coarse-graining.

## 5 Is the coarse-grained process Markov?

Equation (5) is exact *for a Markov chain*. Whether the binned process is one is an empirical question, and we test it four independent ways.



**Figure 1:** The first three tests, per persistence level and arm. Green marks the one cell that passes: silent at  $\rho = 1$ . Test 1 is negative there (a second-order model fits *worse* out of sample), the Chapman–Kolmogorov gap stays near 0.03 out to five steps, and the conditional mutual information is within noise of its permutation null ( $p = 0.15$ ; every other cell gives  $p \approx 0.005$ ).

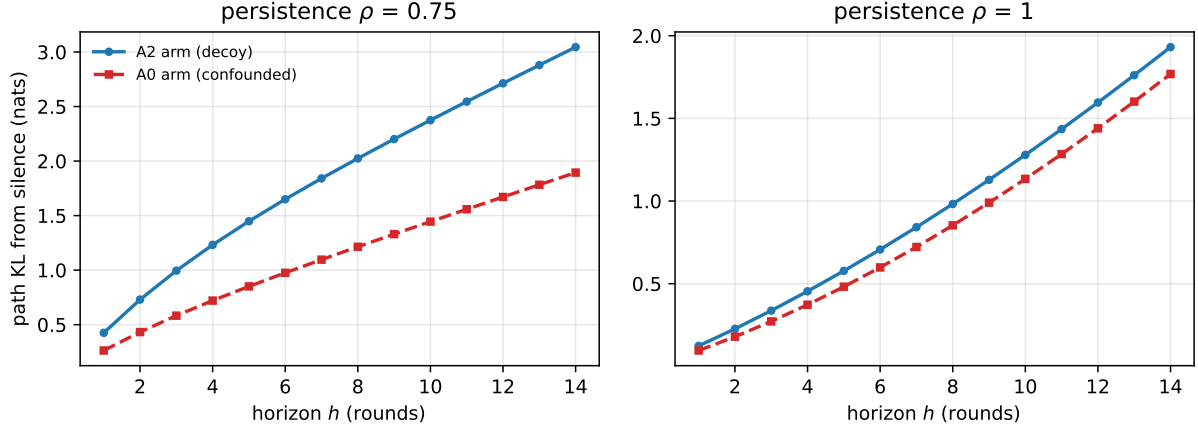
| $\rho$ | arm    | $n$  | order-2 gain   | CK gap ( $n=5$ ) | CMI excess (bits) | $p$          |
|--------|--------|------|----------------|------------------|-------------------|--------------|
| 0.75   | silent | 1800 | 0.0161         | 0.106            | 0.0284            | 0.005        |
| 0.75   | A2     | 5055 | 0.0146         | 0.109            | 0.0261            | 0.005        |
| 0.75   | A0     | 5205 | 0.0139         | 0.110            | 0.0225            | 0.005        |
| 1.00   | silent | 1800 | <b>−0.0050</b> | <b>0.033</b>     | <b>0.0017</b>     | <b>0.154</b> |
| 1.00   | A2     | 2520 | 0.0060         | 0.078            | 0.0130            | 0.005        |
| 1.00   | A0     | 4350 | 0.0025         | 0.068            | 0.0074            | 0.005        |

**Table 1:** All four tests agree. Only  $\rho = 1$  / silent is Markovian in the four-state coordinate. At  $\rho = 0.75$  roughly 0.015 nats per step of memory is unexplained.

1. **Order comparison.** Fit a first-order and a second-order kernel; report the held-out log-likelihood gain per step. A Markov process gains nothing.
2. **Chapman–Kolmogorov.** Compare the empirical  $n$ -step transition matrix with the  $n$ -th matrix power of the one-step kernel, in total variation. A Markov process agrees at all  $n$ .
3. **Conditional independence.** Estimate  $I(s_{t+1}; s_{t-1} \mid s_t)$  and compare it with a permutation null. A Markov process has zero.
4. **Lumpability.** Check whether states that were merged behave identically, by splitting each bin on an extra coordinate and measuring how far the two halves’ outgoing rows differ.

## 5.1 Why lower persistence makes this harder, not easier

The intuition that forgetting destroys memory, and so should *help* Markovianity, is the wrong way round. Forgetting destroys memory in the agents; it does not destroy memory in the *coarse variable*. Two populations with the same average correctness  $e_t$  can differ in how that correctness is distributed — one where every agent knows a little, one where a few agents know a lot. Under perfect memory that difference is frozen and does not affect the next step. Under decay the concentrated population loses more, because its information sits in fewer places, so the two populations move differently from the same value of  $e_t$ . The hidden coordinate is knowledge *spread*, and decay is exactly what makes it matter. The lumpability test confirms it: splitting on spread gives the largest row discrepancy in the silent arm (0.130 at  $\rho = 0.75$ ), while under control the worst offender is instead the survival of decisive facts.



**Figure 2:** Path divergence from silence as a function of horizon. Intervention is *more* displacing at  $\rho = 0.75$  than at  $\rho = 1$  despite fewer posts per episode, consistent with a refresh mechanism: at finite persistence the controller’s facts keep being re-injected into decaying memories, whereas under perfect memory one post is enough and further posts are redundant.

## 6 Information estimates

### 6.1 Quantities

**Path divergence.**  $K = D(P^{\text{ctl}} \| P^{\text{silent}})$  from Eq. (5): how far intervention displaces the trajectory law from silence. Units: nats.

**Intervention information.**  $I(\text{intervention}; s_{0:h})$ , at most 1 bit: can you tell from the trajectory alone whether a controller was running?

**Target information.**  $I(\text{target}; s_{0:h})$ , also at most 1 bit given two targets: can you tell *which way* it was steering? This is the quantity a control-efficiency claim rests on.

**Efficiency.**  $\eta = I(\text{target}; \cdot) / C_\pi$ , where  $C_\pi$  is the control cost measured as the accumulated divergence. Bits of achieved steering per nat of intervention.

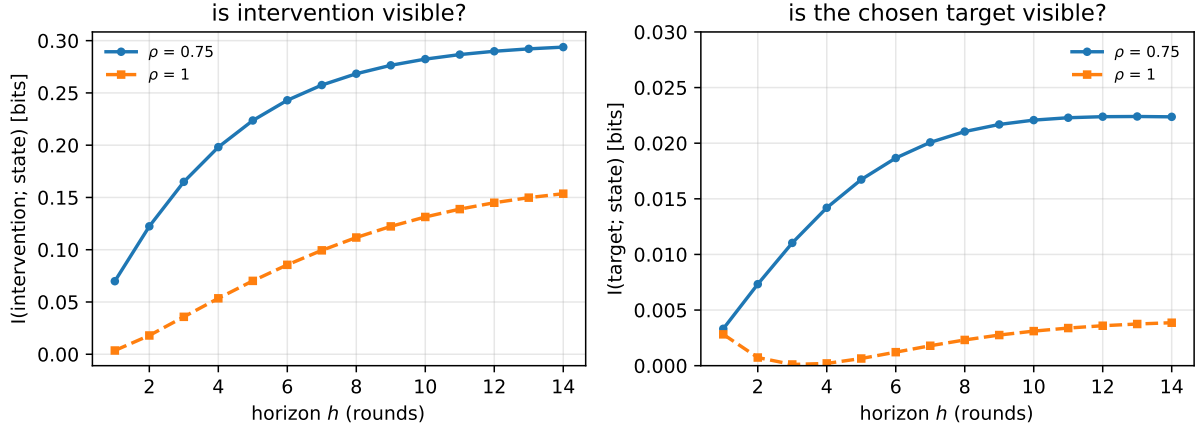
### 6.2 Results at horizon $h = 14$

| $\rho$ | $K(\text{A2} \parallel \text{silent})$ [nats] | $I(\text{interv.}; s)$ [bits] | $I(\text{target}; s)$ [bits] | $\eta$ | posts/episode |
|--------|-----------------------------------------------|-------------------------------|------------------------------|--------|---------------|
| 0.75   | <b>3.045</b> [2.77, 3.28]                     | <b>0.294</b> [0.26, 0.33]     | 0.022                        | 0.006  | 69.1          |
| 1.00   | <b>1.931</b> [1.60, 2.31]                     | <b>0.154</b> [0.12, 0.19]     | 0.004                        | 0.001  | 83.9          |

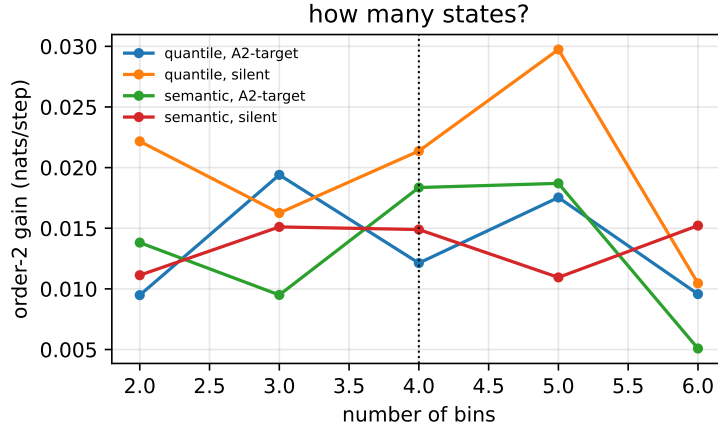
**Table 2:** Bias-corrected cluster-bootstrap intervals (300 resamples, clustered by initialization). Target information and  $\eta$  are *confounded* by the pool defect and are reported for completeness only.

### 6.3 Three warnings about these numbers

**Validity is cell-specific.** Only  $\rho = 1$  / silent passes all four Markov tests. At  $\rho = 0.75$  the numbers carry about 0.015 nats/step of unexplained memory and a Chapman–Kolmogorov gap reaching 0.11 at five steps. Long-horizon path quantities there are indicative, not exact.



**Figure 3:** Left: intervention is clearly visible in the coarse state (0.29 bits of a possible 1 at  $\rho = 0.75$ ). Right: the *direction* of steering is not — note the axis scale. This is the quantitative signature of the pool defect: both arms quote the same facts, so the trajectory cannot tell you which target was configured.



**Figure 4:** Choosing the number of bins. Held-out order-2 gain (lower is more Markovian) against bin count, for quantile and semantic cut points. Dotted line marks the chosen value. Differences among 3, 4 and 5 bins do not survive resampling across seeds.

**EP is not physical dissipation.** The column we compute is the Markov-chain irreversibility  $\sum_{ij} p_i T_{ij} \ln(p_i T_{ij} / p_j T_{ji})$ . No reverse protocol has been justified for this system, and a sequence-reversal score must not be relabelled entropy production. It also *decreases* with horizon here, which is what one expects of a transient chain rather than a stationary one.

**The bootstrap needed a bias correction.** Resampling episodes with replacement duplicates them, which smooths the refitted kernel and pulls any divergence systematically toward zero. Raw percentile intervals therefore excluded their own point estimates. The intervals in Table 2 are shifted by (bootstrap mean — point estimate).

## 7 What this means for the next experiment

1. **Re-run with matched symmetric pools.** Target information cannot be interpreted until the two arms quote different, comparably strong evidence.
2. **Keep  $\rho = 1$  as the primary cell.** It is the only one where the path divergence is exact rather than approximate. Finite persistence remains interesting — it is where intervention

bites hardest — but needs either the 8-state coordinate or substantially more episodes.

3. **Decide the controller’s role deliberately.** It currently holds a pool that can prove nothing (0 of 7,694 minimal proofs), while the agents jointly hold 4. Whether the controller should be able to prove its target is a design choice, not a detail, and it determines whether the experiment measures persuasion or disclosure.
4. **Budget for the estimator, not just the simulation.** The binding constraint on every quantity in Table 2 is transitions per (arm,  $\rho$ ) cell, not episodes.

## Reproducing this

| step                 | command                                 |
|----------------------|-----------------------------------------|
| per-round states     | <code>scripts/build_states.py</code>    |
| epistemic coordinate | <code>scripts/add_coordinates.py</code> |
| the four tests       | <code>scripts/run_four_tests.py</code>  |
| bin sweep            | <code>scripts/bin_sweep.py</code>       |
| estimates            | <code>scripts/estimate.py</code>        |
| this document        | <code>scripts/build_report.py</code>    |

Inputs: the archived study under `agents_control/new_rnd_init_experiment` (read-only) and the generator source in `src/mas_cc/musr_team_allocation_generator`. This analysis lives under `analysis/preanalysis_task003_new_setup_and_coarsegraining/` and is versioned on branch `dev/rsanchez`; its `results/` subdirectory (per-round state files, logs) is not.